

Rodrigo Molina Ávila

Data Scientist / Computational Sociologist

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Profile

Data Scientist and Sociologist, MSc(c) in Data Science from the Faculty of Physical and Mathematical Sciences at the Universidad de Chile, with 5+ years of experience developing analytical and predictive models applied to institutional, academic, and social phenomena. My work lies at the intersection of sociological training —deep understanding of gender, social class, elites, politics, and crisis— and the technical capacity to model them rigorously through knowledge graphs, NLP, machine learning, and data engineering. Specialized in transforming large volumes of heterogeneous data (relational, textual, and network) into actionable knowledge for strategic decision-making. Proven experience designing end-to-end pipelines: from data integration and quality control to executive dashboards and reports for technical and non-technical audiences. Methodological rigor, interdisciplinary communication, and reproducibility as working standards.

Professional Experience

March 2024 – Present **Data Scientist – Applied Research**, *Data and Artificial Intelligence Initiative (IDIA)* — *Millennium Institute for Foundational Research on Data (IMFD)*, Universidad de Chile, Santiago, Chile

- **Designed and ran predictive and clustering models** on the academic trajectories of **3,000+ doctoral fellows funded by the Chilean State**, identifying patterns of scientific production, gender inequality, and social class intersectionality — directly analogous to admission, graduation, and performance analysis.
- **Built robust ETL pipelines** integrating multiple heterogeneous sources (Scopus, ANID, QS/Shanghai rankings, administrative records) with Python and SQL, generating a Neo4j knowledge graph with **over 3,000,000 relationships**, ensuring data quality, consistency, and traceability.
- **Implemented machine learning models** (XGBoost, LightGBM, CatBoost + Optuna) with stratified cross-validation and SHAP interpretability analysis to explain factors associated with fellows' academic success.
- **Designed executive dashboards** (Power BI, D3.js) used in academic publications and briefings for university authorities and ANID policy makers.
- **Technologies:** Python, Neo4j, SQL, scikit-learn, pandas, XGBoost, LightGBM, CatBoost, SHAP, Transformers, Power BI, D3.js, Docker, Git

March 2026 – May 2026 **Consultant — Qualitative Evaluation**, *Data and Artificial Intelligence Initiative (IDIA)*, Professional Master's Program in Artificial Intelligence, FCFM, Santiago, Chile

- **Conducted qualitative evaluation** of relevance, curriculum design, and implementation conditions of the Professional Master's in AI, through a focus group with alumni and interviews with key industry stakeholders.
- **Built a complete analysis pipeline:** speaker diarization, transcription, thematic coding of the corpus, and natural language processing over the structured corpus.
- **Developed a navigable web interface** exposing the full chain of evidence by objective and participant, guaranteeing traceability of the study.
- **Technologies:** Python, audio diarization, NLP, LLMs, HTML/JavaScript

- October 2024 – May 2025 **Research Assistant**, *Center for Social Conflict and Cohesion Studies (COES)*, Chilean Think Tanks in Times of Crisis, Santiago, Chile
- **Developed a data extraction and analysis system** for **20+ institutions**, processing **40,000+ documents** with automated web scraping and NLP pipelines, published on **GitHub Pages**.
 - **Modeled influence networks** using topic analysis (BERTopic) and network analysis, identifying production patterns and temporal trends.
 - **Trained the research team** in R, Python, and Git for reproducible data analysis.
 - **In progress**: academic paper in computational sociology journals.
 - **Technologies**: Python, R, SQL, Web Scraping, pandas, scikit-learn, GitHub Pages, D3.js, Power BI
- September 2024 – May 2025 **Research Assistant**, *FONDECYT 11241304*, Inequalities in STEM Academia, Santiago, Chile
- **Designed a mixed-methods approach** to analyze academic trajectories with a gender and social class intersectional perspective on a population of **3,000+ researchers**.
 - **Implemented clustering models** (K-Means, hierarchical clustering) and multiple correspondence analysis (MCA) for academic profile segmentation.
 - **Produced interactive visualizations** to communicate complex findings to interdisciplinary teams and non-technical stakeholders.
 - **Technologies**: Python, R, pandas, scikit-learn, applied statistics, Power BI
- February 2023 – March 2024 **Research Assistant**, *FONDECYT POSTDOC 3210579*, Think Tanks amid Presidential Transitions, Santiago, Chile
- **Built a unified database** with **48,000+ publications** through automated extraction from APIs and official websites, integrating multi-source data.
 - **Designed a relational model** and transformation pipelines ensuring data consistency and availability for longitudinal analysis.
 - **Developed Power BI dashboards** for time-series production density analysis.
 - **Technologies**: Python, R, SQL, Power BI, REST APIs
- February 2022 – 2023 **Research Assistant**, *Mini COES — Constitutional Convention*, Expert networks: migration and decentralization, Santiago, Chile
- **Built databases** and data mining on expert networks in the migration and decentralization debate.
 - **Ethnography** and network generation inside the former National Congress.
 - **Semi-structured interviews**, transcription, and coding in Atlas.ti.
 - **Writing** and publication production.
- September 2021 – February 2022 **Technical Staff**, *FONDECYT Regular 11200602*, Geographic Inequality in a Double Register, Santiago, Chile
- **Systematization of categories** for qualitative discourse analysis in the Antofagasta and La Araucanía markets.
 - **Grounded Theory**: theoretical analysis, transcription, and categorization.
- May 2018 – September 2018 **Professional Internship**, *Gendarmería de Chile*, Classification and Segmentation Department, National Directorate, Santiago, Chile
- **Methodological update** of the classification form: database analysis, statistical tests, and instrument design.
 - **Fieldwork** inside Santiago 1 and CCP Puente Alto prisons: qualitative sampling and bibliographic analysis.
- 2019 **Survey Taker**, *Datavoz*, Barómetro de la Política, Santiago, Chile
- Data collection for the Barómetro de la Política survey.
- May 2025 **Quantitative Consultant**, *Fab Consulting*, Research Valorization Program Evaluation, ANID, Santiago, Chile
- Technical proposal for the Undersecretariat of Science, Technology, Knowledge and Innovation.
 - Quantitative analysis of program results.

Education

- March 2024 – Present **MSc in Data Science**, *Universidad de Chile*, Faculty of Physical and Mathematical Sciences (FCFM), Santiago, Chile
- **Specialization:** Machine Learning, Knowledge Graphs, NLP, Causal Inference, Relational and Non-Relational Databases, Artificial Intelligence
 - **Thesis approved (defense pending):** Neo4j Knowledge Graph modeling the academic trajectories and scientific production of **3,143 ANID doctoral fellows** (2014–2016 cohorts) with BERTopic and Graph Data Science. Public platform: interrogandolaacademia.cl
- 2014 – 2019 **BA in Sociology (Sociologist)**, *Universidad Andrés Bello*, Faculty of Education and Social Sciences, Santiago, Chile
- **Specialization:** Quantitative and Qualitative Research, Applied Statistics, Public Policy Evaluation, Organizational Development

Technical Skills

Languages	Python, R, SQL (PostgreSQL, SQLite), Cypher (Neo4j), SPARQL
ML & Statistics	scikit-learn, XGBoost, LightGBM, CatBoost, SHAP, Optuna, statsmodels, Clustering, Dimensionality Reduction, Predictive Modeling, Cross-Validation, Community Detection (Louvain, Leiden), Centrality Measures (PageRank, Betweenness)
Data Engineering	pandas, NumPy, ETL Pipelines, Web Scraping, REST APIs, Multi-source Integration, Data Quality Control
Databases	SQL (PostgreSQL, SQLite), NoSQL, Neo4j, Property Graphs, RDF/OWL, Wikidata, Excel Ecosystem
NLP	Transformers, BERTopic, SBERT, Embeddings, Sentiment Analysis, spaCy, NLTK, Speaker Diarization, LLM-Assisted Qualitative Analysis
Viz & BI	Power BI, D3.js, Plotly, Matplotlib, Seaborn, GitHub Pages, HTML/CSS/JavaScript
MLOps & Deploy	MLflow, Docker, Git, Apache Airflow, Jupyter, Gradio
Harness Engineering & AI APIs	Claude Code, omp, MCP, AI APIs (Anthropic, OpenAI), RAG pipelines, prompt engineering, local LLMs (Ollama), fine-tuning / LoRA

Publications

- 2026 **Untangling the Academic Network: Analyzing Scientific Trajectories with Graphs and Natural Language Processing**, *AMW 2026 (CEUR-WS)*, Under review
- **Co-author:** Sebastián Ferrada Aliaga (IDIA – IMFD). Knowledge graph of academic trajectories combining topic modeling (BERTopic) and community detection (Louvain, Leiden).
- 2026 **Think Tanks and Political Crises in Chile, 2011–2022; or, The Crystallization of the Intellectual Infrastructure in a Middle-Income Country**, *Latin American Research Review (Cambridge University Press)*, M. González Hernando
- **Acknowledged researcher** for contributions to the Chilean think tank research line (FONDECYT POSTDOC 3210579 / COES). doi:10.1017/lar.2026.10142

Teaching

- March 2026 – July 2026 **Teaching Assistant — Knowledge Graphs**, *MSc in Data Science, FCFM*, Universidad de Chile, Santiago, Chile
- Teaching support in property graph modeling, Cypher queries, and Neo4j Graph Data Science algorithms.
- August 2026 – Present **Teaching Assistant — Information Retrieval (IR)**, *FCFM Diploma Program*, Universidad de Chile, Santiago, Chile
- Teaching support in information retrieval fundamentals: indexing, ranking, and search system evaluation.

- March 2026 – **Teaching Assistant — Fundamentals of Sociology**, *Universidad Diego Portales*, Faculty of Sociology, Santiago, Chile
- Teaching support in the Fundamentals of Sociology course.
- August 2025 – June 2026 **Co-supervisor — Thesis Project**, *Department of Computer Science (DCC)*, FCFM, Universidad de Chile, Santiago, Chile
- Co-supervisor of Matías Vega's thesis: **Web platform for access to data on science funding in Chile**.
 - Result: the public platform **interrogandolaacademia.cl**.
- Second Semester 2022 **Teaching Assistant — Latin American Politics**, *Faculty of Government, Universidad de Chile*, Political Science (Dr. Luis Garrido Vergara), Santiago, Chile
- Teaching support in the Latin American Politics course.
- Second Semester 2022 **Teaching Assistant — Human Behavior in Organizations**, *Faculty of Government, Universidad de Chile*, Political Science (Dr. Luis Garrido Vergara), Santiago, Chile
- Teaching support in the Human Behavior in Organizations course.
- Second Semester 2022 **Teaching Assistant — Empirical Analysis of Public Policies**, *Faculty of Government, Universidad de Chile*, Public Administration (Dr. Juan Carlos Cortázar Velarde), Santiago, Chile
- Teaching support in the Empirical Analysis of Public Policies course.

Conferences

- August 2026 **IV FCFM Graduate Symposium**, *Trajectories of ANID Fellows through Knowledge Graphs and Natural Language Processing*, Universidad de Chile, Santiago, Chile
- **Co-author with Sebastián Ferrada (IDIA – IMFD)**: what public investment in doctoral training produces and how the scientific community that sustains it is structured.
- November 2025 **Research Findings Presentation**, *Chilean Think Tanks in Turbulent Times: 2019–2023*, Universidad Alberto Hurtado, Santiago, Chile
- **Speaker**: MiniCOES research findings with an interdisciplinary team. **View presentation**
- October 2025 **XII COES International Conference**, *Is Democracy Really the Problem?*, COES, Santiago, Chile
- **Presenter**: “Expert knowledge in turbulent times: think tanks during Chile's political crises (2019–2023)”
- November 2022 **IX COES International Conference**, *The Future of the Commons*, COES, Santiago, Chile
- **Presenter**: “The Role of Expert Knowledge in the Constitutional Convention”

Languages

Spanish	Native
English	Advanced (academic reading, writing, and conversation)
Chinese	Basic